

Matteo Rufolo

PHD RESEARCHER IN META-LEARNING · MACHINE LEARNING RESEARCHER

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Summary

PhD researcher (IDSIA, USI-SUPSI) building domain-agnostic Transformer and state-space architectures for in-context learning, meta-learning, and bilevel optimization on sequential data. My research centers on fast adaptation under distribution shift and amortized inference, the same primitives underlying agentic and few-shot LLM behavior, applied to dynamical systems within a broader scientific machine learning framework. I have authored 10 papers, all built on heavy JAX/PyTorch GPU-parallelized training, that goes from methodological contribution analyzing the learning framework and proposing new methods to application scenarios like real robot arm or real battery analysis. Previously at CERN, I engineered large-scale nonlinear physics simulation pipelines and solved ill-conditioned inverse problems. I'm looking for a research internship applying meta-learning and physics-informed AI to scientific and industrial systems, from physical dynamics to time-series forecasting and multimodal foundation model, working on real physical simulation pipelines and use cases.

Technical Skills

Languages	Python (Advanced), MATLAB (Advanced), C (Intermediate), Julia (Intermediate)
DL Frameworks	PyTorch, JAX, Hugging Face (Datasets), Scikit-learn, PySR,
MLOps & Tools	Docker, Weights & Biases, GitHub/GitLab,
HPC/GPU	CUDA, distributed training, GPU parallelization, Monte Carlo simulations, Multi-GPU distributed training (single-node, up to 8 GPUs)
Core Competencies	Meta-Learning, Bilevel Optimization, In-Context Learning, Sequential Modeling, Transformers, State-Space Models, Efficient Neural Architectures, Distributed GPU Training, Speed-up Learning, Scientific ML, Reliable ML Systems, Interpretability, State Estimation

Work Experience

IDSIA, SUPSI-USI

Lugano, Switzerland

PHD RESEARCHER IN META-LEARNING

Jan. 2024 - Present

- Developing meta-learning architectures for sequential state and parameter estimation in dynamical systems: Transformer-based in-context learners that adapt at inference time from short, noisy trajectory observations; latent dynamics representations that encode system-specific behavior in low-dimensional manifolds; and diffusion-based generative sequence models for robot dynamics prediction
- Built and trained reduced-order meta-models in JAX on multi-GPU single-node setups, matching full-model performance with significantly reduced compute and only 10% of the training data, direct experience in compute-efficient model training and evaluation
- Authored and co-authored 10 research papers (7 accepted in IEEE L-CSS, CDC, ECC, Automatica; 3 under review in Automatica, T-RL, Neurocomputing) focusing on robust meta-learning, representation learning, and fast adaptation for sequential dynamical data.
- Collaborating with industrial partners to deploy hardware-aware, GPU-optimized meta-learning pipelines for real-time inference and spatio-temporal prediction tasks.

European Organization for Nuclear Research (CERN)

Meyrin, Switzerland

COMPUTATIONAL RESEARCHER (TECHNICAL STUDENT)

Sep. 2022 - Oct. 2023

- Engineered a high-performance Python framework for large-scale nonlinear physical simulation (beam-beam dynamics, LHC luminosity optimization), achieving two orders of magnitude speedup via GPU parallelization and algorithmic scaling, directly applicable to computationally intensive Earth system simulation pipelines.
- Processed thousands of collision datasets with Pandas, NumPy, and SciPy, resolving ill-conditioned inverse problems via advanced numerical techniques, large-scale data curation and evaluation at production scale.
- Developed and validated nonlinear physics models using Monte Carlo simulation and closed-form analytical methods (q-Gaussian distributions, quadrupole integrals), demonstrating deep familiarity with physical simulation systems and stochastic modeling.
- Published methodologies in the CERN Document Server and JACoW IPAC paper, culminating in a Master's thesis awarded 110/110 cum laude.

Basque Center of Applied Mathematics (BCAM)

Bilbao, Spain

MACHINE LEARNING RESEARCHER (INTERNSHIP)

May 2022 - Aug. 2022

- Developed and deployed a novel MinMax Risk Classification algorithm in MATLAB, leveraging the Best Linear Unbiased Estimator (BLUE) of the sample mean.
- Trained, benchmarked, and optimized diverse ML architectures (Neural Networks, SVMs, Decision Trees) to minimize classification error against Root Mean Square Error (RMSE).
- Established rigorous version control and reproducible data analysis workflows using GitHub for the research team.

Publications

Accepted

- Marco Forgione, Ankush Chakrabarty, Dario Piga, Matteo Rufolo, Alberto Bemporad (2026). "Manifold meta-learning for reduced-complexity neural system identification". In **Automatica** (Q1)
- Moroncelli, Rufolo et al. (2026). 'Diffusion Sequence Models for Generative In-Context Meta-Learning of Robot Dynamics.' In **IEEE Control and Decision Conference 26**
- Matteo Rufolo et al.(2026). 'Variational meta-learning inference for low-dimensional neural system identification.' In **IEEE Control and Decision Conference 2026**
- Matteo Rufolo, Dario Piga, Marco Forgione (2026) "Amortized Neural-Symbolic Inverse Dynamics: Robust and Invariant Parameter Inference via Latent Dynamics". In **AI4Physics Workshop International Conference on Machine Learning 26**
- Matteo Rufolo, Dario Piga, Marco Forgione (2025). "Distributionally robust minimization in meta-learning for system identification". In **IEEE Control Systems Letters** (Q1) **Finalist in IEEE CSS Swiss Chapter Young Author Best Journal Paper Award 2026**
- Matteo Rufolo, Dario Piga, Gabriele Maroni, Marco Forgione (2025). "Enhanced Transformer architecture for in-context learning of dynamical systems". In **European Control Conference 25**
- Dario Piga, Matteo Rufolo, Gabriele Maroni, Manas Mejari, Marco Forgione (2024). "Synthetic data generation for system identification: leveraging knowledge transfer from similar systems". In **IEEE Control and Decision Conference 24**
- Matteo Rufolo, Lorenzo Giacometti, Andrea Fornara, Guido Sterbini, Eleanor Lamb (2024). "JACOW: Numerical methods for emittance computation from luminosity". In **JACoW International Particle Accelerator Conference 24**

Under review / Submitted

- Matteo Rufolo, Dario Piga, Filippo Pura, Marco Forgione (2025). "In-Context Learning for System Identification: Unified Framework, Enhanced Methods, and Benchmark Results". In revision to **Automatica** (Q1)
- Matteo Rufolo, Francois Germain, Gordon Wichern, Alberto Bemporad, Dario Piga, Marco Forgione, Ankush Chakrabaty (2025). "Neural Amortization for Nonlinear System Identification". In revision to **Neurocomputing** (Q1)
- Charbel Barsomian, Matteo Rufolo, Andrea Monguzzi, Navvab Kashiri, Francesco Braghini, Marco Forgione, Loris Roveda (2025). "A Physics-Informed In-Context Learning Framework for Online Force Prediction in Contact Tasks". In revision to **IEEE Transaction of Robot Learning**

Education

Polytechnic University of Turin

M.S. IN MATHEMATICAL ENGINEERING

Turin, Italy

Sep. 2020 - Oct. 2023

- Thesis in collaboration with CERN: Numerical Methods for the Optimization of the LHC Luminosity Computation
- Final grade 110/110 cum laude
- Curriculum included: Computational Fluid Dynamics, Continuum Mechanics, Fluid Mechanics, PDEs, Transport Models and Kinetic Theories, Nonlinear Systems, providing strong mathematical foundations in physical simulation.

Polytechnic University of Turin

B.S. IN MATHEMATICS FOR ENGINEERING

Turin, Italy

Sep. 2017 - Sep. 2020

- Final grade 110/110

Training & Schools

ProbAI school

5 DAYS

Vilnius, Lithuania

Aug. 2026

Model Predictive Control

5 DAYS

IMT Lucca, Italy

Mar. 2025 - Apr. 2025

16th Machine Learning and Advanced Statistics Summer School

5 DAYS

Universidad Politécnica de Madrid,

Spain

Jun. 2024

Spring School on Data-driven Model Learning for Dynamic Systems

5 DAYS

Ecole Centrale de Lyon, France

Apr. 2024

Introduction to Accelerator Physics (CERN Accelerator School)

14 DAYS

Kaunas, Lithuania

Sep. 2022 - Oct. 2022

Language skills

Italian

Native

English

Fluent (working language for research, publication and international collaboration).

Research interest

Meta-learning · Fast adaptation · In-context learning · State Estimation · Uncertainty Quantification · DNN interpretability · Reliable ML · Sequential modeling · Diffusion Models · Generative Sequence Modeling